

Properly colored paths in the union of two Hamiltonian paths can have length $O(\sqrt{n})$

Alejandro Lizardi*

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Abstract

For a permutation σ of $\{0, 1, \dots, n-1\}$, let H_σ be the two-edge-colored multigraph on $\{0, \dots, n-1\}$ whose blue edges form the natural-order Hamiltonian path and whose red edges form the Hamiltonian path induced by the value order of σ , and let $\rho(\sigma)$ denote the maximum number of vertices on a properly colored path of H_σ (a simple path whose consecutive edges alternate in color). Every two-edge-colored multigraph that is the union of two Hamiltonian paths on a common vertex set arises this way. We prove that $\rho_{\min}(n) := \min_\sigma \rho(\sigma) = O(\sqrt{n})$; explicitly, $\rho_{\min}(n) \leq 8\sqrt{n} + 20$ for all $n \geq 100$. Consequently $\rho_{\min}(n)/n \rightarrow 0$: there is no constant $c > 0$ such that the union of two Hamiltonian paths must contain a properly colored path on cn vertices. This answers, in the negative, the main question posed in the author's earlier note [1], where a positive answer was proposed as a route toward improved bounds for Lovász's problem on Hamiltonian paths in vertex-transitive graphs; we emphasize that the present result closes only that particular route, not the underlying matching-edge question (Section 1.2).

The proof introduces an inflation (substitution) operation on permutations together with a per-permutation invariant, the *transit number* $M(\pi)$, which bounds how many vertices a properly colored path can collect while passing through an inflated copy of π . Our main lemma is a complete classification of the constrained terminal-to-terminal paths (*admissible visits*) of an explicit gadget family: for every even $b \geq 6$, the permutation $T_b = (0, b-1, 2, 3, \dots, b-2, 1)$ admits exactly five admissible visits, of sizes 1, 2, 2, 3, 3; in particular $M(T_b) = 3$, independent of b . The classification turns on a parity obstruction in a long run of doubled edges. For the resulting explicit permutations $\sigma_{a,b}$ on $n = ab$ vertices we prove the two-sided bounds $2a + 2b - 7 \leq \rho(\sigma_{a,b}) \leq 4a + 2b$, so the family provably realizes the order $\sqrt{n}$ when $a \asymp b$. Whether $\rho_{\min}(n) = o(n)$ can be complemented by any superconstant lower bound remains open: even $\rho_{\min}(n) \rightarrow \infty$ is not known. A second, independent construction with the same conclusion is given in an appendix, and all computations are reproducible from the ancillary code and logs.

*Independent researcher. The proofs and computations in this paper were AI-assisted by Claude, under the Periapsis research structure. All computational claims are reproducible from the ancillary files accompanying this submission; see Section 7.



1 Introduction

Let S_n denote the set of bijections $\sigma: \{0, 1, \dots, n-1\} \rightarrow \{0, 1, \dots, n-1\}$. Following [1], associate with $\sigma \in S_n$ the two-edge-colored multigraph H_σ on vertex set $\{0, 1, \dots, n-1\}$ with

- **P-edges** (“position”, blue): $\{i, i+1\}$ for $0 \leq i \leq n-2$;
- **V-edges** (“value”, red): $\{\sigma^{-1}(v), \sigma^{-1}(v+1)\}$ for $0 \leq v \leq n-2$.

Each color class is a Hamiltonian path. A vertex pair may carry both a P-edge and a V-edge; these are retained as *parallel edges of different colors* (“doubled pairs”). Conversely, the union of any two Hamiltonian paths on a common vertex set, with one path blue and the other red, is color-isomorphic to H_σ for some σ (identify the vertex set with $\{0, \dots, n-1\}$ along the blue path, and let σ rank the vertices along the red path).

A *properly colored path* (PC path) is a simple path—all vertices distinct—whose consecutive edges have different colors. A single vertex is a PC path. Let $\rho(\sigma)$ be the maximum number of vertices on a PC path in H_σ , and

$$\rho_{\min}(n) = \min_{\sigma \in S_n} \rho(\sigma).$$

The note [1], an earlier study by the present author, developed a local calculus for PC paths in H_σ , determined the extremal value of ρ on a family of block-reversal permutations, refuted a half-bound conjecture at $n = 18$, and exhibited stacked families with ρ/n tending (conjecturally) to $5/11$ and then $10/28$. The main question it left open, stated there in self-contained form, was:

Does there exist $c > 0$ such that every two-edge-colored multigraph that is the union of two Hamiltonian paths on the same vertex set contains a properly colored simple path on at least cn vertices?

The original motivation comes from a relaxation chain described in [1]: a linear lower bound on $\rho_{\min}(n)$ would transfer, through an exact conversion lemma, to a linear lower bound on a matching-edge parameter $f(n)$ for a family of subcubic graphs arising in the Norin–Steiner–Thomassé–Wollan approach [3] to Lovász’s 1969 question [2] on Hamiltonian paths in connected vertex-transitive graphs. We stress the scope of that motivation: the relaxation was proposed in [1], not in the broader literature, and the present paper’s negative answer closes only this relaxation (see Section 1.2).

Beyond that specific motivation, the question is a natural extremal problem about alternating paths. Properly colored (“alternating”) paths and cycles in edge-colored graphs have been studied extensively since the survey of Bang-Jensen and Gutin [5]; see also [6, 7]. The positive results in that line are sufficient conditions—typically large color degree or dense host graphs—for long PC paths or cycles to exist. The graphs H_σ sit at the opposite end of the spectrum: every vertex has at most two edges of each color, and the question is extremal rather than Dirac-type. We are not aware of prior work on the union of two Hamiltonian paths besides [1].

1.1 Results

We answer the question in the negative.



Theorem 1.1 (Main Theorem). $\rho_{\min}(n) = O(\sqrt{n})$. Explicitly, $\rho_{\min}(n) \leq 8\sqrt{n} + 20$ for all $n \geq 100$. Consequently $\rho_{\min}(n)/n \rightarrow 0$.

The construction is explicit. For even $b \geq 6$ let

$$T_b = (0, b-1, 2, 3, \dots, b-2, 1) \in S_b$$

be the identity permutation with the values 1 and $b-1$ exchanged, and for $a \geq 2$ define $\sigma_{a,b} \in S_{ab}$ by

$$\sigma_{a,b}(jb + r) = jb + T_b(r) \quad (0 \leq j < a, 0 \leq r < b),$$

i.e. a consecutive copies of T_b on consecutive value ranges. We prove the two-sided estimate

$$2a + 2b - 7 \leq \rho(\sigma_{a,b}) \leq 4a + 2b \quad (a \geq 2, b \geq 6 \text{ even}), \quad (1)$$

the upper bound by the inflation calculus below (Theorem 6.1, Proposition 6.2), the lower bound by an explicit path (Proposition 6.3). Thus $\rho(\sigma_{a,b}) = \Theta(a + b)$: with $a \asymp b \asymp \sqrt{n}$ the family realizes the order $\sqrt{n}$ exactly, so the upper bound is tight for this family and not an artifact of a construction whose true value is smaller. (For $a = 2$ both bounds in (1) are weak: indeed $\rho(\sigma_{2,b}) \leq n = 2b$ trivially. The construction is informative for $a \geq 3$, and asymptotically for $a \rightarrow \infty$.) Exact machine computation, reproducible from the ancillary files, gives $\rho(\sigma_{a,b}) = 2a + 2b - 4$ at every tested point up to $n = 720$ (Section 7); we use this exact formula nowhere in the proofs and state it only as data.

The proof of the upper bounds has two independent components.

1. An inflation calculus (Sections 3–4). Call a set B of consecutive positions whose σ -image is a set of consecutive values an *interval block*. A short argument (Lemma 3.2) shows that at most *four* edges of H_σ join B to its complement, and that they attach at four canonical *terminal* vertices of B , with prescribed colors. If $\sigma = \tau[\pi_1, \dots, \pi_a]$ is built by inflation (each position block an interval block with pattern π_i , arranged by an outer permutation τ), then any PC path decomposes into at most $2a - 1$ *visits* to blocks, and each visit other than the first and last must enter and leave through terminals subject to color constraints. Defining the *transit number* $M(\pi)$ as the largest number of vertices such a constrained passage (*admissible visit*) through H_π can collect, we obtain (Theorem 4.4)

$$\rho(\tau[\pi_1, \dots, \pi_a]) \leq (2a - 3) \max_i M(\pi_i) + 2 \max_i |\pi_i|.$$

2. A classification of the gadget’s admissible visits (Section 5). For every even $b \geq 6$ we determine the admissible visits of H_{T_b} *completely* (Theorem 5.4): up to reversal there are exactly five, of sizes 1, 2, 2, 3, 3. In particular

$$M(T_b) = 3,$$

independent of b . The mechanism is a parity obstruction: H_{T_b} contains a long run of doubled edges which any terminal-to-terminal passage must either avoid or cross completely, and a full crossing forces the color of the departing edge precisely so that, for even b , every continuation is blocked. For odd b the parity reverses and a transit can collect all b vertices; exact computation confirms $M(T_b) = b$ for odd b . The even/odd contrast shows that the parity, not some feature of the argument, is what bounds the transit number.



Combining the two components and balancing a against b yields Theorem 1.1; the crude bound $\rho(\sigma_{a,b}) \leq 6a + 2b$, which needs only the value $M(T_b) = 3$ and not the full classification, already gives $\rho_{\min}(n) \leq 10\sqrt{n} + 10$, and the refined constant $8\sqrt{n} + 20$ uses the classification once more (Proposition 6.2). The proofs are elementary and self-contained; the computations reported in Section 7 are independent verifications, not ingredients of the proofs.

Appendix A contains a *second, independent* construction proving $\rho_{\min}(n) = O(\sqrt{n})$: chains of reversed odd blocks separated by singletons, where a different parity trap blocks every crossing, with its own complete proof. The two constructions share the parity-trap idea but are different families with different edge inventories, and they were found and verified independently, by different methods and different solvers. We give both because a redundant proof of the main claim seems worth the space.

Section 8 isolates what is known about lower bounds, including the machine-verified refutation of a natural conjectured inequality ($\rho \geq$ number of ± 1 -blocks) that holds exhaustively through $n = 12$ but fails at $n = 240$.

1.2 Scope: what this does and does not say about Lovász’s problem

The conversion lemma of [1] maps PC paths in H_σ to matching-edge-rich paths in an associated subcubic graph $G(n, \sigma)$ *one-directionally*. Our construction makes properly colored paths short; it says nothing about paths in $G(n, \sigma)$ that are allowed to use runs of same-colored edges. The known bound $f(n) = \Omega(n^{4/5})$ for the matching-edge parameter, which rests on the circumference theorem for 3-connected cubic graphs of Liu, Yu and Zhang [4] as used in [3], is untouched, and $f(n) = \Theta(n)$ remains entirely possible. What dies is only the hope, raised in [1], of proving $f(n) = \Omega(n)$ via the properly-colored relaxation.

On the lower-bound side our knowledge is thin: we prove no superconstant lower bound on $\rho_{\min}(n)$, and to our knowledge even $\rho_{\min}(n) \rightarrow \infty$ is open. The quantities controlling the natural approaches are discussed in Section 8, where we also state the conjecture $\rho_{\min}(n) = \Theta(\sqrt{n})$ — at present an upper bound plus structural circumstantial evidence, no more.

2 Preliminaries

Throughout, $\sigma \in S_n$ and H_σ are as in Section 1; colors are P and V, and for a color c we write $\bar{c}$ for the other color. Paths are sequences of distinct vertices $u_1, \dots, u_k$ ($k \geq 1$) together with, for each consecutive pair, a chosen edge (with its color) joining them; in a doubled pair the path commits to one of the two parallel edges. A path is *properly colored* (PC) if consecutive edges have different colors. The *size* of a path is its number of vertices k . Since each vertex pair carries at most one edge of each color and paths have distinct vertices, a path uses pairwise distinct edges, even in the multigraph sense.

Observation 2.1. A two-edge-colored multigraph is the union of a blue and a red Hamiltonian path on a common vertex set of size n if and only if it is color-isomorphic to H_σ for some $\sigma \in S_n$. Hence Theorem 1.1 is equivalent to its abstract formulation.

3 Interval blocks and ports

Definition 3.1 (Interval block, pattern). An *interval block* of $\sigma \in S_n$ is a set of positions $B = \{p_\ell, p_\ell + 1, \dots, p_r\}$ whose image is a set of consecutive values $\sigma(B) = \{w_\ell, \dots, w_r\}$. Its



pattern is the permutation $\pi \in S_b$, $b = |B|$, defined by $\pi(t) = \sigma(p_\ell + t) - w_\ell$.

Lemma 3.2 (Port structure). *Let B be an interval block of σ with positions $[p_\ell, p_r]$, values $[w_\ell, w_r]$, and pattern π . Then:*

- (a) *The sub-multigraph of H_σ induced on B is color-isomorphic to H_π under the map $p \mapsto p - p_\ell$.*
- (b) *Every edge of H_σ with exactly one endpoint in B is one of at most four: the P-edges $\{p_\ell - 1, p_\ell\}$ and $\{p_r, p_r + 1\}$ (when they exist), whose endpoints in B are p_ℓ and p_r ; and the V-edges $\{\sigma^{-1}(w_\ell - 1), \sigma^{-1}(w_\ell)\}$ and $\{\sigma^{-1}(w_r), \sigma^{-1}(w_r + 1)\}$ (when they exist), whose endpoints in B are $\sigma^{-1}(w_\ell)$ and $\sigma^{-1}(w_r)$.*

Proof. (a) P-edges internal to B are the pairs $\{p, p + 1\} \subseteq [p_\ell, p_r]$, which correspond exactly to the P-edges of H_π . A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v + 1)\}$ has both endpoints in B if and only if both $v, v + 1 \in [w_\ell, w_r]$ (as $\sigma(B) = [w_\ell, w_r]$ and σ is a bijection), and these edges correspond exactly to the V-edges of H_π under the stated map, since $\pi^{-1}(v - w_\ell) = \sigma^{-1}(v) - p_\ell$ for $v \in [w_\ell, w_r]$. Doubled pairs match doubled pairs, so the isomorphism preserves the multigraph structure with colors.

(b) A P-edge $\{p, p + 1\}$ with exactly one endpoint in the interval $[p_\ell, p_r]$ satisfies $p + 1 = p_\ell$ or $p = p_r$. A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v + 1)\}$ with exactly one endpoint in B has exactly one of $v, v + 1$ in $[w_\ell, w_r]$, forcing $v + 1 = w_\ell$ or $v = w_r$. $\square$

Definition 3.3 (Terminals, roles, port colors). Let $\pi \in S_b$, $b \geq 2$. The *terminals* of H_π are the four *roles*

$$p_0 = \text{vertex } 0, \quad p_1 = \text{vertex } b - 1, \quad v_0 = \text{vertex } \pi^{-1}(0), \quad v_1 = \text{vertex } \pi^{-1}(b - 1),$$

with *port colors* $\text{pc}(p_0) = \text{pc}(p_1) = \text{P}$ and $\text{pc}(v_0) = \text{pc}(v_1) = \text{V}$. Distinct roles may occupy the same vertex (a role is a label, not a vertex); since $b \geq 2$, two roles with the *same* port color always occupy distinct vertices.

By Lemma 3.2, in any interval block with pattern π the at most four outgoing edges attach (inside the block) precisely at the terminals, and the color of an outgoing edge equals the port color of the role at which it attaches.

Definition 3.4 (Admissible visit, transit number). Let $\pi \in S_b$, $b \geq 2$. An *admissible visit* of H_π is either

- (i) a PC path $u_1, \dots, u_k$ in H_π with $k \geq 2$, together with a choice of roles r at u_1 and r' at u_k , such that the first edge has color $\neq \text{pc}(r)$ and the last edge has color $\neq \text{pc}(r')$; or
- (ii) a single vertex carrying two distinct roles r, r' with $\text{pc}(r) \neq \text{pc}(r')$.

The *transit number* $M(\pi)$ is the maximum size of an admissible visit, and $M(\pi) = 0$ if none exists. For $b \leq 1$ set $M(\pi) = 1$.

Since a classification of admissible visits must account for the role choices, we record two trivial but load-bearing facts.



Lemma 3.5 (Reversal; endpoint constraint). (a) If $(u_1, \dots, u_k; r, r')$ is an admissible visit, so is the reversed $(u_k, \dots, u_1; r', r)$. Hence admissible visits may be classified as undirected vertex sequences with unordered role pairs. (b) In a type-(i) admissible visit, both u_1 and u_k are terminal vertices, the roles r, r' are distinct, and with two colors the conditions read: first edge color $= \overline{\text{pc}(r)}$, last edge color $= \overline{\text{pc}(r')}$.

Proof. (a) Reversal swaps first and last edges and the two endpoint constraints. (b) The roles occupy $u_1 \neq u_k$, hence are distinct; the rest is the definition. $\square$

4 Inflation and the Inflation Lemma

Definition 4.1 (Inflation). Let $a \geq 2$, let τ be a permutation of $\{1, \dots, a\}$, and let $\pi_1, \dots, \pi_a$ be permutations of sizes $b_1, \dots, b_a \geq 2$, $N = \sum_i b_i$. Put $s_i = \sum_{j < i} b_j$ (position offsets) and $t_i = \sum_{j: \tau(j) < \tau(i)} b_j$ (value offsets). The *inflation* $\sigma = \tau[\pi_1, \dots, \pi_a] \in S_N$ is defined by

$$\sigma(s_i + r) = t_i + \pi_i(r), \quad i = 1, \dots, a, \quad 0 \leq r < b_i.$$

The value intervals $[t_i, t_i + b_i - 1]$ partition $\{0, \dots, N - 1\}$, so σ is a bijection. Each position block $B_i = [s_i, s_i + b_i - 1]$ is an interval block of σ with value interval $[t_i, t_i + b_i - 1]$ and pattern π_i , so Lemma 3.2 applies to all blocks simultaneously. We call edges of H_σ with endpoints in different blocks *cross edges*.

Lemma 4.2. $H_{\tau[\pi_1, \dots, \pi_a]}$ has exactly $2(a - 1)$ cross edges, counted with multiplicity: $a - 1$ P-colored (one per consecutive pair of position blocks) and $a - 1$ V-colored (one per consecutive pair of value intervals). Each block is incident to at most 4 cross edges, attaching at its terminals as in Lemma 3.2(b), and each terminal role of each block carries at most one cross edge.

Proof. A P-edge $\{p, p + 1\}$ is a cross edge iff $p + 1 \in \{s_2, \dots, s_a\}$: $a - 1$ edges. A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v + 1)\}$ is a cross edge iff $v, v + 1$ lie in different value intervals, which happens for exactly the $a - 1$ values v ending a value interval. The attachment statement is Lemma 3.2(b); per block, each of the four port positions (P at each end of the position interval, V at each end of the value interval) is met by at most one cross edge. $\square$

Lemma 4.3 (Visit decomposition). Let $\sigma = \tau[\pi_1, \dots, \pi_a]$, $a \geq 2$, and let Q be a PC path in H_σ . Partition the vertex sequence of Q into maximal runs of consecutive vertices lying in a common block; call these runs visits. Then:

- (a) Consecutive visits are joined by exactly one cross edge; distinct consecutive pairs use distinct cross edges. Hence if Q has q visits, then $q - 1 \leq 2(a - 1)$, i.e. $q \leq 2a - 1$.
- (b) Every visit that is neither the first nor the last is an admissible visit of the copy of H_{π_i} on its block; in particular it has at most $M(\pi_i)$ vertices.
- (c) The first and the last visit have at most $\max_i b_i$ vertices each.

Proof. (a) The edge of Q joining the last vertex of one visit to the first vertex of the next has endpoints in different blocks (by maximality of runs), so it is a cross edge. Q has distinct vertices, hence pairwise distinct edges, so the $q - 1$ connecting edges are distinct cross edges; by Lemma 4.2 there are $2(a - 1)$ of them in all.



(b) Let $W = u_1, \dots, u_k$ be an interior visit in block B_i . There is an *entry* cross edge e_{in} of Q arriving at u_1 and an *exit* cross edge e_{out} leaving u_k . By Lemma 4.2, e_{in} attaches at a terminal role r at u_1 and has color $\text{pc}(r)$; likewise e_{out} at a role r' at u_k with color $\text{pc}(r')$. All edges of Q between vertices of W have both endpoints in B_i , so under Lemma 3.2(a) the run W is a PC path in the copy of H_{π_i} . If $k \geq 2$: the PC condition of Q at u_1 (between e_{in} and the first edge of W) forces the first edge of W to have color $\neq \text{pc}(r)$, and similarly at u_k ; thus W , with the roles r, r' , is admissible of type (i). If $k = 1$: the PC condition at u_1 between e_{in} and e_{out} forces $\text{pc}(r) \neq \text{pc}(r')$, so u_1 carries two roles of different port colors: admissible of type (ii).

(c) A visit lies in one block, so its size is at most that block's size. $\square$

Theorem 4.4 (Inflation Lemma). *For $a \geq 2$,*

$$\rho(\tau[\pi_1, \dots, \pi_a]) \leq (2a - 3) \max_i M(\pi_i) + 2 \max_i b_i.$$

Proof. Let Q attain ρ and let q be its number of visits. If $q = 1$, then $\rho \leq \max_i b_i$ and the bound holds. If $q \geq 2$, then by Lemma 4.3 the $q - 2 \leq 2a - 3$ interior visits have at most $\max_i M(\pi_i)$ vertices each and the two end visits at most $\max_i b_i$ each; visits partition the vertices of Q . $\square$

Theorem 4.4 holds for *every* outer permutation τ ; no property of τ is used. The problem is thereby reduced to finding patterns with small transit number relative to their size.

5 The gadget: classification of admissible visits of H_{T_b}

Definition 5.1. For even $b \geq 6$ let $T_b \in S_b$ be given by

$$T_b(0) = 0, \quad T_b(1) = b - 1, \quad T_b(i) = i \quad (2 \leq i \leq b - 2), \quad T_b(b - 1) = 1.$$

Thus T_b^{-1} fixes 0 and each $v \in [2, b - 2]$, and maps $1 \mapsto b - 1$, $b - 1 \mapsto 1$ (T_b is an involution). We first record the complete edge inventory; recall $b \geq 6$.

Lemma 5.2 (Inventory). *The edges of H_{T_b} are exactly:*

- doubled pairs $\{i, i + 1\}$ for $2 \leq i \leq b - 3$ (each carrying one P - and one V -edge);
- P -only edges $\{0, 1\}$, $\{1, 2\}$, $\{b - 2, b - 1\}$;
- V -only edges $\{0, b - 1\}$, $\{2, b - 1\}$, $\{1, b - 2\}$.

Consequently the full edge lists at the relevant vertices are:

vertex 0 :	$P\{0, 1\}; V\{0, b - 1\}$ (degree 2),
vertex 1 :	$P\{0, 1\}, P\{1, 2\}; V\{1, b - 2\}$ (unique V -edge),
vertex $b - 1$:	$P\{b - 2, b - 1\}; V\{0, b - 1\}, V\{2, b - 1\}$ (unique P -edge),
vertex 2 :	$P\{1, 2\};$ doubled $\{2, 3\}; V\{2, b - 1\}$,
vertex $b - 2$:	doubled $\{b - 3, b - 2\}; P\{b - 2, b - 1\}; V\{1, b - 2\}$,
vertex i , $3 \leq i \leq b - 3$:	doubled $\{i - 1, i\}$ and $\{i, i + 1\}$ only.

The terminals are $p_0 = v_0 = \text{vertex } 0$, $v_1 = \text{vertex } 1$, $p_1 = \text{vertex } b - 1$.



Proof. P-edges are $\{i, i+1\}$, $0 \leq i \leq b-2$, by definition. The V-edges, indexed by $v = 0, \dots, b-2$, are $\{T_b^{-1}(v), T_b^{-1}(v+1)\}$:

$$v = 0 : \{0, b-1\}; \quad v = 1 : \{b-1, 2\}; \quad 2 \leq v \leq b-3 : \{v, v+1\}; \quad v = b-2 : \{b-2, 1\}.$$

A pair $\{i, i+1\}$ is doubled iff it occurs in both lists: this happens exactly for $2 \leq i \leq b-3$. (For $b \geq 6$: $\{0, 1\}$ would require $|T_b(1) - T_b(0)| = b-1 = 1$, false; $\{1, 2\}$ requires $b-3 = 1$, false; $\{b-2, b-1\}$ requires $b-3 = 1$, false. The three long V-edges join non-adjacent positions since $b-1, b-3 \geq 3$.) The per-vertex lists and the terminals ($T_b^{-1}(0) = 0$, $T_b^{-1}(b-1) = 1$) follow by inspection. $\square$

Let $S = \{2, 3, \dots, b-2\}$ (the *segment*), so $|S| = b-3$. By Lemma 5.2, the sub-multigraph induced on S is the path $2-3-\dots-(b-2)$ with every pair doubled; the only edges from S to its complement are the two at vertex 2 (P $\{1, 2\}$, V $\{2, b-1\}$) and the two at vertex $b-2$ (P $\{b-2, b-1\}$, V $\{1, b-2\}$); and no terminal lies in S .

Lemma 5.3 (Segment behavior). *Let Q be a PC path in H_{T_b} .*

- (a) *While inside S , Q moves monotonically in position (each interior vertex of S is adjacent only to its two position-neighbors), and Q can leave S only from vertex 2 or vertex $b-2$.*
- (b) *Suppose Q enters S at one end (2 or $b-2$) by an edge of color c , at a moment when no vertex of S has yet been visited, and later leaves S . Then either it leaves immediately from the entry vertex (using the other external edge there, which has color $\bar{c}$), or it traverses all of S to the far end; in the latter case its $b-4$ internal edges alternate colors starting with $\bar{c}$, so (since b is even, making $b-4$ even) the last internal edge has color c , and any edge by which Q then leaves S has color $\bar{c}$.*

Proof. (a) is immediate from the inventory: a simple path cannot reverse inside a path graph. For (b): leaving from the entry vertex means the run in S is that single vertex; the two external edges at 2 (P $\{1, 2\}$, V $\{2, b-1\}$) have different colors, likewise at $b-2$, and the PC condition forces the exit edge to differ from the entry edge, i.e. to be the other one, of color $\bar{c}$. Otherwise the run proceeds monotonically and can only exit at the far end, having covered all of S with internal edges $g_1, \dots, g_{b-4}$. The PC condition at the entry vertex gives $g_1 = \bar{c}$ (both colors are available on every doubled pair; the PC condition is what forces the alternation), alternation gives $g_m = \bar{c}$ for odd m and $g_m = c$ for even m , and $b-4$ is even, so $g_{b-4} = c$; the PC condition at the far end forces the exit edge to have color $\neq c$. $\square$

5.1 The classification theorem

Theorem 5.4 (Classification of admissible visits). *Let $b \geq 6$ be even. Up to reversal (Lemma 3.5), the admissible visits of H_{T_b} are exactly the following five, listed with their (unique) role pairs:*

	vertices	edges (colors)	role pair
(V1)	0	—	$\{p_0, v_0\}$
(V2)	0, $b-1$	V $\{0, b-1\}$	$\{p_0, p_1\}$
(V3)	0, 1	P $\{0, 1\}$	$\{v_0, v_1\}$
(V4)	$b-1$, 0, 1	V $\{0, b-1\}$, P $\{0, 1\}$	$\{p_1, v_1\}$
(V5)	$b-1$, 2, 1	V $\{2, b-1\}$, P $\{1, 2\}$	$\{p_1, v_1\}$



In particular: the role pairs $\{p_0, v_1\}$ and $\{v_0, p_1\}$ admit no visit; the only visit for $\{p_0, p_1\}$ is (V2); the only visit for $\{v_0, v_1\}$ is (V3); and $M(T_b) = 3$.

Proof. That each listed visit is admissible is a direct check from Lemma 5.2. (V1): vertex 0 carries p_0 and v_0 with different port colors. (V2): first = last edge $V \neq \text{pc}(p_0) = \text{pc}(p_1) = P$. (V3): first = last edge $P \neq \text{pc}(v_0) = \text{pc}(v_1) = V$. (V4) and (V5): first edge $V \neq \text{pc}(p_1)$, last edge $P \neq \text{pc}(v_1)$, and the middle vertex makes the colors alternate. The role pairs listed are the only possible ones: e.g. in (V2) the first edge is V, so the role at 0 must have port color $\neq V$, i.e. p_0 ; the same one-line check determines the role pair of each visit.

For completeness we must show no other admissible visit exists. Type-(ii) visits need a vertex carrying two roles of different port colors; by Lemma 5.2 the only vertex carrying two roles is $0 = p_0 = v_0$, giving (V1). A type-(i) visit starts at a terminal vertex with a constrained first color (Lemma 3.5(b)). The terminal vertices are $0, 1, b-1$, and the roles they carry leave exactly four possible (start vertex, first color) *stems*:

- (A) $(0, V)$ [role p_0], (B) $(0, P)$ [role v_0], (C) $(1, P)$ [role v_1], (D) $(b-1, V)$ [role p_1].

Every admissible visit, read from its own first vertex, extends one of these stems, so it suffices to enumerate, for each stem, all PC continuations, and to record those moments at which the path stands at a terminal vertex with a last-edge color admissible for a role there. Throughout we use the *forcing facts* from Lemma 5.2: (F1) vertex 0 has exactly one P-edge ($\{0, 1\}$) and one V-edge ($\{0, b-1\}$); (F2) vertex 1 has exactly one V-edge ($\{1, b-2\}$); (F3) vertex $b-1$ has exactly one P-edge ($\{b-2, b-1\}$); and the segment behavior of Lemma 5.3. Recall that no vertex of S is a terminal, so a path currently stranded in S can never end an admissible visit there.

Stem (A): start 0, first edge V. By (F1) the first edge is $V\{0, b-1\}$; the path stands at $b-1$ with last color V. *Record:* role p_1 at $b-1$ has $\text{pc} = P \neq V$, so $(0, b-1)$ is admissible: this is (V2). To continue, the next edge must be $\neq V$, hence by (F3) it is $P\{b-2, b-1\}$; the path enters S at $b-2$ with entry color P and S untouched. By Lemma 5.3(b) there are two ways out of S :

- *Immediate exit at $b-2$ by $V\{1, b-2\}$:* the path stands at 1 with last color V. The only role at 1 is v_1 with $\text{pc} = V$: *not* admissible. Continuing from 1: next $\neq V$, so a P-edge: $\{0, 1\}$ is excluded (0 visited), leaving $P\{1, 2\}$ into S at 2. From 2, next $\neq P$: $V\{2, b-1\}$ is excluded ($b-1$ visited), leaving the V-side of $\{2, 3\}$. From 3 onward the path is inside S and moves monotonically upward (Lemma 5.3(a)); it can never exit, because the only exit vertices are 2 (behind it) and $b-2$ (already visited), so it strands in S . No admissible visit arises.
- *Full crossing to 2:* the path arrives at 2 by an internal edge of color P (Lemma 5.3(b) with $c = P$), and any exit edge must have color V. The only V-colored external edge at 2 is $\{2, b-1\}$, whose endpoint $b-1$ is visited. The path strands in S (vertex 2 is in S); no admissible visit arises.

Stem (A) yields exactly (V2).

Stem (B): start 0, first edge P. By (F1) the first edge is $P\{0, 1\}$; the path stands at 1 with last color P. *Record:* role v_1 at 1 has $\text{pc} = V \neq P$: admissible: this is (V3). To continue: next $\neq P$, so by (F2) it is $V\{1, b-2\}$; the path enters S at $b-2$ with entry color V, S untouched. By Lemma 5.3(b):



- *Immediate exit at $b - 2$ by $P\{b - 2, b - 1\}$:* the path stands at $b - 1$ with last color P . The only role at $b - 1$ is p_1 with $pc = P$: not admissible. Continuing from $b - 1$: $next \neq P$, so a V -edge: $\{0, b - 1\}$ is excluded (0 visited), leaving $V\{2, b - 1\}$ into S at 2. From 2, $next \neq V$: $P\{1, 2\}$ is excluded (1 visited), leaving the P -side of $\{2, 3\}$; the path moves monotonically upward inside S and strands ($b - 2$ visited). No admissible visit.
- *Full crossing to 2:* arrival color V , so any exit edge must be P ; the only P -colored external edge at 2 is $\{1, 2\}$, and 1 is visited. Strands. No admissible visit.

Stem (B) yields exactly (V3).

Stem (C): start 1, first edge P . The P -edges at 1 are $\{0, 1\}$ and $\{1, 2\}$ (Lemma 5.2); two branches.

- *Branch $1 \rightarrow 0$.* At 0, last color P . *Record:* roles at 0: p_0 ($pc = P$, fails), v_0 ($pc = V$, admissible): this is (V3) reversed. Continue: $next \neq P$, by (F1) it is $V\{0, b - 1\}$; at $b - 1$, last color V . *Record:* role p_1 : admissible: this is (V4) reversed. Continue: $next \neq V$, by (F3) $P\{b - 2, b - 1\}$ into S at $b - 2$, entry color P , S untouched. Immediate exit by $V\{1, b - 2\}$: 1 visited. Full crossing: arrival at 2 with color P , exit must be V : only $\{2, b - 1\}$, visited. Strands; branch closed.
- *Branch $1 \rightarrow 2$.* The path enters S at 2 by $P\{1, 2\}$, entry color P , S untouched. Immediate exit by $V\{2, b - 1\}$: at $b - 1$, last color V . *Record:* role p_1 : admissible: this is (V5) reversed. Continue from $b - 1$: $next \neq V$, by (F3) $P\{b - 2, b - 1\}$ into S at $b - 2$ — but now S is touched (vertex 2): from $b - 2$ the non- P continuations are $V\{1, b - 2\}$ (1 visited) and the V -side of $\{b - 3, b - 2\}$, after which the path moves monotonically down inside S and strands at 3 (2 visited). Branch closed. Alternatively, instead of exiting S immediately at 2, the path may cross: full crossing $2 \rightarrow b - 2$ with entry color P arrives at $b - 2$ by color P ; exit must be V : only $\{1, b - 2\}$, and 1 is visited. Strands; branch closed.

Stem (C) yields exactly (V3), (V4), (V5) (reversed).

Stem (D): start $b - 1$, first edge V . The V -edges at $b - 1$ are $\{0, b - 1\}$ and $\{2, b - 1\}$; two branches.

- *Branch $b - 1 \rightarrow 0$.* At 0, last color V . *Record:* roles at 0: v_0 fails, p_0 admissible: this is (V2) reversed. Continue: $next \neq V$, by (F1) $P\{0, 1\}$; at 1, last color P . *Record:* role v_1 : admissible: this is (V4). Continue: $next \neq P$, by (F2) $V\{1, b - 2\}$ into S at $b - 2$, entry color V , S untouched. Immediate exit by $P\{b - 2, b - 1\}$: $b - 1$ visited. Full crossing: arrival at 2 with color V , exit must be P : only $\{1, 2\}$, and 1 is visited. Strands; branch closed.
- *Branch $b - 1 \rightarrow 2$.* The path enters S at 2 by $V\{2, b - 1\}$, entry color V , S untouched. Immediate exit by $P\{1, 2\}$: at 1, last color P . *Record:* role v_1 : admissible: this is (V5). Continue from 1: $next \neq P$, by (F2) $V\{1, b - 2\}$ into S at $b - 2$ — S touched (vertex 2): from $b - 2$ the non- V continuations are $P\{b - 2, b - 1\}$ ($b - 1$ visited) and the P -side of $\{b - 3, b - 2\}$, then monotone descent stranding at 3 (2 visited). Branch closed. Alternatively, full crossing $2 \rightarrow b - 2$ with entry color V arrives by color V ; exit must be P : only $\{b - 2, b - 1\}$, and $b - 1$ is visited. Strands; branch closed.



Stem (D) yields exactly (V2), (V4), (V5).

Collecting the four stems: every admissible visit is one of (V1)–(V5) up to reversal, and each is realized only with the listed role pair (checked at each *Record* step). In particular no admissible visit realizes $\{p_0, v_1\}$ or $\{v_0, p_1\}$, and the maximum size is 3, attained by (V4) and (V5). $\square$

Corollary 5.5. $M(T_b) = 3$ for every even $b \geq 6$, and $c^* := \liminf_n \rho_{\min}(n)/n$ satisfies $c^* \leq 6/b$ for every even $b \geq 6$; hence $c^* = 0$ and the displayed question of Section 1 has a negative answer.

Proof. $M(T_b) = 3$ is part of Theorem 5.4. For the second claim, take $\pi_i = T_b$ for all i in Theorem 4.4 and let $a \rightarrow \infty$: $\rho_{\min}(ab)/(ab) \leq ((2a-3) \cdot 3 + 2b)/(ab) \rightarrow 6/b$. $\square$

Remark 5.6 (The parity is essential). For odd b , $b-4$ is odd and the last internal edge of a full crossing has color $\bar{c}$, so the exit edge may have color c . The route $0 \xrightarrow{P} 1 \xrightarrow{V} b-2 \rightsquigarrow 2 \xrightarrow{V\{2,b-1\}} b-1$ (full crossing in the middle) then becomes properly colored and is an admissible visit for $\{v_0, p_1\}$ of size b . Exact computation (Section 7) confirms $M(T_b) = b$ for odd $b \in \{7, 9, 11, 13\}$ and $M(T_b) = 3$ for even $6 \leq b \leq 100$, and confirms the *exact list* of Theorem 5.4 for even $6 \leq b \leq 14$ by independent brute-force enumeration of all admissible visits (ancillary file `verify_M.py`).

Corollary 5.7 (Two visits to one T_b -block). Let $\sigma = \tau[\pi_1, \dots, \pi_a]$ be an inflation, let B_i be a block whose pattern π_i equals T_b for some even $b \geq 6$, and let Q be a PC path in H_σ . Then the interior visits of Q to B_i have total size at most 4.

Proof. By Lemma 4.2, the block has at most four incident cross edges, one per role; each interior visit uses two of them (its entry and its exit, distinct edges), and every cross edge is used by Q at most once. Hence there are at most two interior visits; if there is at most one, its size is at most $M(T_b) = 3 \leq 4$. If there are two, they use all four cross edges, so their role pairs partition $\{p_0, p_1, v_0, v_1\}$, and the two visits are vertex-disjoint (disjoint runs of the simple path Q). By the classification (Theorem 5.4) the three possible partitions give:

- $\{p_0, p_1\} / \{v_0, v_1\}$: the only visits are (V2) = $(0, b-1)$ and (V3) = $(0, 1)$, which share vertex 0 — impossible;
- $\{p_0, v_1\} / \{v_0, p_1\}$: no visits exist for either pair — impossible;
- $\{p_0, v_0\} / \{p_1, v_1\}$: sizes 1 (V1) and at most 3 (V4 or V5): total at most 4.

$\square$

6 Proof of the Main Theorem

Theorem 6.1. (a) For every even $b \geq 6$ and $a \geq 2$, $\rho(\sigma_{a,b}) \leq 6a + 2b$, where $\sigma_{a,b} = \text{id}_a[T_b, \dots, T_b]$, i.e. $\sigma_{a,b}(jb + r) = jb + T_b(r)$.

(b) For every $n \geq 100$, $\rho_{\min}(n) \leq 10\sqrt{n} + 10$.

In particular $\rho_{\min}(n) = O(\sqrt{n})$ and $\rho_{\min}(n)/n \rightarrow 0$, using only $M(T_b) = 3$ and the Inflation Lemma.



Proof. (a) Immediate from Theorem 4.4 with $M(\pi_i) = 3$ and $b_i = b$: $\rho \leq (2a-3) \cdot 3 + 2b \leq 6a + 2b$.

(b) Let $b = 2\lfloor \sqrt{n}/2 \rfloor$, so b is even, $\sqrt{n} - 2 < b \leq \sqrt{n}$, and $b \geq 10$ since $n \geq 100$. Let $a = \lfloor n/b \rfloor \geq 2$ and $s = n - (a-1)b$, so $b \leq s < 2b$. Define $\sigma \in S_n$ as the inflation (any outer permutation, say the identity) of the following list of blocks:

- if s is even: $a-1$ copies of T_b and one copy of T_s (valid: s even, $s \geq b \geq 10$);
- if s is odd: $a-1$ copies of T_b , one copy of T_{s-3} , and one block of size 3 with the identity pattern $\iota_3 = (0, 1, 2)$ (valid: $s-3$ is even and $s-3 \geq b-2 \geq 8$, since $s \geq b$ with s odd and b even force $s \geq b+1$).

Every block has transit number at most 3: the T -blocks by Theorem 5.4, the single ι_3 -block trivially (it has only 3 vertices). The number of blocks is $m \leq a+1 \leq n/b+1$ and the largest block has size at most $2b-1$. By Theorem 4.4,

$$\rho_{\min}(n) \leq (2m-3) \cdot 3 + 2(2b-1) \leq \frac{6n}{b} + 4b - 5.$$

Since $b \geq \sqrt{n} - 2$ and $\sqrt{n} \geq 10$,

$$\frac{6n}{b} \leq \frac{6n}{\sqrt{n}-2} = 6\sqrt{n} + \frac{12\sqrt{n}}{\sqrt{n}-2} \leq 6\sqrt{n} + 15,$$

using $\sqrt{n}/(\sqrt{n}-2) \leq 10/8$ for $\sqrt{n} \geq 10$. With $4b \leq 4\sqrt{n}$ we get $\rho_{\min}(n) \leq 10\sqrt{n} + 10$. $\square$

The refined constants use the classification once more, through Corollary 5.7.

Proposition 6.2 (Refined constants). *For every even $b \geq 6$ and $a \geq 2$, $\rho(\sigma_{a,b}) \leq 4a + 2b$; and for every $n \geq 100$, $\rho_{\min}(n) \leq 8\sqrt{n} + 20$.*

Proof. First claim: let Q be a PC path in $H_{\sigma_{a,b}}$ with q visits. If $q = 1$ then $|Q| \leq b$. Otherwise, by Lemma 4.3 the first and last visits have at most b vertices each, and by Corollary 5.7 the interior visits contribute at most 4 per block, i.e. at most $4a$ in total. Hence $\rho \leq 4a + 2b$.

Second claim: take the construction from the proof of Theorem 6.1(b). Interior visits contribute at most 4 per T -block (Corollary 5.7; the corollary applies verbatim to T_b , T_s , and T_{s-3}) and at most $2 \cdot 3 = 6$ in total for the single ι_3 -block (at most two interior visits — Lemma 4.2 — of size at most 3 each). End visits contribute at most $2(2b-1)$. Hence

$$\rho_{\min}(n) \leq 4\left(\frac{n}{b} + 1\right) + 6 + 2(2b-1) = \frac{4n}{b} + 4b + 8 \leq (4\sqrt{n} + 10) + 4\sqrt{n} + 8 \leq 8\sqrt{n} + 20,$$

using $4n/(\sqrt{n}-2) = 4\sqrt{n} + 8\sqrt{n}/(\sqrt{n}-2) \leq 4\sqrt{n} + 10$ for $\sqrt{n} \geq 10$. $\square$

Proposition 6.3 (Explicit long path: the family is $\Theta(a+b)$). *For every even $b \geq 6$ and $a \geq 2$, $\rho(\sigma_{a,b}) \geq 2a + 2b - 7$.*

Proof. Write (j, ℓ) for the vertex at position $jb + \ell$ ($0 \leq j < a$, $0 \leq \ell < b$), so block j is a copy of H_{T_b} on $\{(j, 0), \dots, (j, b-1)\}$, and the values are: $(j, 0) \mapsto jb$, $(j, 1) \mapsto jb + b - 1$, $(j, \ell) \mapsto jb + \ell$ for $2 \leq \ell \leq b-2$, and $(j, b-1) \mapsto jb + 1$. Besides the internal edges (Lemma 5.2 in each block), the cross edges are: the P-edges $\{(j, b-1), (j+1, 0)\}$ (consecutive positions) and the V-edges $\{(j, 1), (j+1, 0)\}$ (values $jb + b - 1$ and $(j+1)b$). Within block j , note the V-edge $\{(j, 0), (j, b-1)\}$ (values jb and $jb + 1$) and the V-edge $\{(j, 2), (j, b-1)\}$ (values $jb + 2$ and $jb + 1$).

The following walk is a PC path; colors are indicated over each arrow, and each edge exists by the inventory just listed.



- *Stage 1 (block 0).* Start at $(0, b - 2)$ and descend the segment to $(0, 2)$ along the doubled run, choosing colors $V, P, V, \dots$ alternately; this uses $b - 4$ edges, and since $b - 4$ is even the last of them (into $(0, 2)$) has color P . Then take $V \{(0, 2), (0, b - 1)\}$, then the P -cross edge to $(1, 0)$. Colors: $\dots, P, V, P$ — properly colored.
- *Stage 2 (blocks $1, \dots, a - 2$).* From $(j, 0)$ (just entered by a P -cross edge) take $V \{(j, 0), (j, b - 1)\}$, then the P -cross edge to $(j + 1, 0)$; repeat. Colors alternate $P, V, P, \dots$ throughout.
- *Stage 3 (block $a - 1$).* From $(a - 1, 0)$ (entered by a P -cross edge) take $V \{(a - 1, 0), (a - 1, b - 1)\}$, then $P \{(a - 1, b - 2), (a - 1, b - 1)\}$, then descend the segment from $(a - 1, b - 2)$ to $(a - 1, 2)$ choosing colors $V, P, \dots$ alternately, and stop at $(a - 1, 2)$.

All vertices are distinct (stages use disjoint blocks; within each stage the listed vertices are distinct), and consecutive edge colors differ at every junction as indicated. The vertex count is $(b - 2)$ in stage 1 ($b - 3$ segment vertices plus $(0, b - 1)$), $2(a - 2)$ in stage 2, and $(b - 1)$ in stage 3 ($(a - 1, 0)$, $(a - 1, b - 1)$, and the $b - 3$ segment vertices), for a total of $2a + 2b - 7$. The construction is verified mechanically, edge by edge, in the ancillary file `verify_path.py`. $\square$

Remark 6.4 (Constants; exact values). Together, Propositions 6.2 and 6.3 give $2(a + b) - 7 \leq \rho(\sigma_{a,b}) \leq 4a + 2b$: the family is $\Theta(a + b)$, and with $a \asymp b$ it realizes $\Theta(\sqrt{n})$, so the upper bound of Theorem 1.1 is achieved by this family up to the constant. Exact machine computation (Section 7) gives $\rho(\sigma_{a,b}) = 2a + 2b - 4$ at every tested point up to $n = 720$; we conjecture this holds for all $a \geq 3$, even $b \geq 6$, but we neither prove nor use it. Optimizing the proven bounds at $n = ab$: the upper bound $4a + 2b$ with $b \asymp \sqrt{2n}$ gives $\rho_{\min}(n) \leq 4\sqrt{2n} + O(1) \approx 5.66\sqrt{n}$ along such n .

7 Computational verification

None of the proofs above depend on computation. This section specifies the computations that independently verify them, and the empirical sharpness of the construction. **All code, command lines, and output logs are provided as ancillary files with this submission** (directory `artifact/`; file `README.md` there maps each claim below to a script and a log; a manifest with SHA-256 checksums is included).

Algorithm. The basic object is the exact longest-PC-path search. Its state space is the set of triples (*visited set*, *current vertex*, *last edge color*); the initial states are all $(\{x, w\}, w, c)$ over edges $\{x, w\}$ of color c ; a transition extends a state by an unvisited neighbor along an edge of the other color. Every PC path corresponds to a unique state, so $\rho(\sigma)$ is the maximum cardinality of the visited set over all *reachable* states, computed by an exhaustive search that visits every reachable state exactly once (a memo set of states; no pruning of any kind). The method is exact by construction; its cost is the number of reachable states. That number is exponential in general — on unstructured instances we used the method only for $n \leq 19$ — but is small on the structured families here, because the gadget strands most partial paths quickly: for $\sigma_{a,b}$ we measure 57,066 reachable states at $n = 80$, 5,012,186 at $n = 600$, and 7,273,586 at $n = 720$ (the logs report the count for every table row). The constrained variants computing $M(\pi)$ (fixed start/end terminals and first/last colors) are the same search with the initial and accepting states restricted.



Implementations. Three independent implementations were used: a plain recursive enumerator and a memoized iterative one (both Python, in the artifact), which agree on 300 random permutations of sizes 4–8; and a separately written C++ solver (also in the artifact) from a parallel verification effort, which agrees with the Python solvers on every instance checked by both, including exact values of $\rho(\sigma_{a,b})$ at $(a, b) \in \{(6, 8), (5, 10), (4, 12)\}$ and the trap-family values of Appendix A at six points. In addition, the artifact contains a *minimal independent verifier* (`verify_M.py`, under 100 lines, written directly from Definition 3.4 with no shared code), which re-derives $M(T_b)$ and the complete list of admissible visits by brute force for $b \leq 14$.

Verified statements. Each item states the claim, its range, and its log.

1. **Anchors from [1]** (`validate2.log`, `search_small.log`): $\rho(\sigma_m) = 2m + 3$ for the block-reversal family at $m = 1, 2, 3, 4$ ($n = 7, 11, 15, 19$); $\rho(\sigma^*) = 10$ for the $n = 18$ counterexample of [1]; $\rho_{\min}(7) = 5$ and $\rho_{\min}(8) = 6$ by full enumeration of S_7, S_8 .
2. **Classification** (`verify_M.log`): for even $b \in \{6, 8, 10, 12, 14\}$, the admissible visits of H_{T_b} are *exactly* (V1)–(V5) of Theorem 5.4; for odd $b \in \{7, 9, 11\}$, $M(T_b) = b$.
3. **The family at scale** (`check_all_even.log`): $M(T_b) = 3$ for *every* even $6 \leq b \leq 100$; $M(T_b) = b$ for odd $b \in \{7, 9, 11, 13\}$ (`validate2.log`).
4. **Optimality of the gadget at small sizes** (`search_small.log`): full enumeration of S_b for $4 \leq b \leq 9$ gives $\min_{\pi \in S_b} M(\pi) = 3$ in every case; no gadget of size ≤ 9 beats transit number 3.
5. **Explicit lower-bound path** (`verify_path.log`): the path of Proposition 6.3 is checked edge-by-edge at seven (a, b) points including $(20, 20)$.
6. **Exact values** (`endtoend.log`, `bigtable.log`): for $\sigma_{a,b}$:

a	b	n	ρ (exact)	$2a+2b-4$	$4a+2b$	ρ/n
10	8	80	32	32	56	0.400
12	8	96	36	36	64	0.375
12	12	144	44	44	72	0.306
16	12	192	52	52	88	0.271
20	12	240	60	60	104	0.250
16	16	256	60	60	96	0.234
24	16	384	76	76	128	0.198
20	20	400	76	76	120	0.190
30	16	480	88	88	152	0.183
24	20	480	84	84	136	0.175
30	20	600	96	96	160	0.160
36	20	720	108	108	184	0.150

Every exact value equals $2a + 2b - 4$ and sits inside the proven window $[2a+2b-7, 4a+2b]$ of (1); the two rows at $n = 480$ (differing in ρ) illustrate that $\rho(\sigma_{a,b})$ tracks a and b separately, not merely their product. At $n = 720$ the ratio 0.150 is far below the smallest ratio previously conjectured to be asymptotically attainable, $10/28 \approx 0.357$ [1].



7. **Exhaustive small- n statistics** (`logs/exh7.txt–exh12.txt`, produced by the C++ solver): the joint distribution of $(b(\sigma), \rho(\sigma))$ over all of S_n for $7 \leq n \leq 12$, used in Section 8; at $n = 12$ this is an enumeration of all 479,001,600 permutations.
8. **Landscape** (exploratory; `probe.py`): under adjacent transpositions, $\sigma_{4,6}$ ($n = 24$, $\rho = 16$) is a strict local minimum of ρ , while greedy descent from random starts stalls around $\rho \approx 23$ — consistent with the flat-landscape observations of [1]; extremal permutations of this type are found by design, not by local search.

Items 1–7 are exhaustive within their stated ranges and reproducible by running the listed scripts; item 8 is exploratory.

8 Lower bounds and open problems

Following [1], a ± 1 -block of σ is a maximal interval of positions on which consecutive values differ by a fixed $\varepsilon \in \{+1, -1\}$ (singletons allowed); let $b(\sigma)$ be their number and $s_{\max}(\sigma)$ the largest block size. Two elementary facts frame the lower-bound question. First, the ± 1 -blocks partition the positions, so

$$s_{\max}(\sigma) \cdot b(\sigma) \geq n, \quad \text{hence} \quad \max(s_{\max}(\sigma), b(\sigma)) \geq \sqrt{n}.$$

Second, $\rho(\sigma) \geq s_{\max}(\sigma)$ always (traverse a largest block along its doubled run; [1]). Consequently, if there were an absolute $c > 0$ with $\rho(\sigma) \geq c b(\sigma)$ for all σ , then $\rho_{\min}(n) \geq c\sqrt{n}$ would follow, and with Theorem 6.1:

Conjecture 8.1. $\rho_{\min}(n) = \Theta(\sqrt{n})$.

We emphasize the asymmetry of the evidence: the upper bound is proved, while for the lower bound we have only the following partial information about ρ versus $b(\sigma)$, which we believe is of independent interest.

- **Exhaustively for $n \leq 12$:** $\rho(\sigma) \geq b(\sigma)$ for every $\sigma \in S_n$ (machine-verified over all 479,001,600 permutations at $n = 12$; artifact logs `exh7.txt–exh12.txt`; equality occurs).
- **Asymptotically the constant 1 fails:** each copy of T_b contributes four ± 1 -blocks ($\{0\}$, $\{1\}$, $[2, b-2]$, $\{b-1\}$), so $b(\sigma_{a,b}) = 4a$, while exact machine computation (`bigtable.log`) gives $\rho(\sigma_{20,12}) = 60 < 80 = b(\sigma_{20,12})$ at $n = 240$, and similarly $76 < 96$ at $(24, 16)$ and $96 < 120$ at $(30, 20)$. These are finite, reproducible counterexamples to $\rho \geq b(\sigma)$.
- **Proven window:** by (1), $\frac{\rho(\sigma_{a,b})}{b(\sigma_{a,b})} \in [\frac{1}{2} + \frac{2b-7}{4a}, 1 + \frac{b}{2a}]$; the conjectured exact formula $2a + 2b - 4$ would give limit $\frac{1}{2}$ as $a/b \rightarrow \infty$. We know of no σ with $\rho(\sigma) < \frac{1}{2}b(\sigma)$.

Thus the sharp form of the lower-bound question is whether $\rho \geq c b(\sigma)$ holds for some $c \leq \frac{1}{2}$; any such c settles Conjecture 8.1. At the opposite extreme, even $\rho_{\min}(n) \rightarrow \infty$ appears to be open: the bound $\rho \geq s_{\max} + 1$ of [1] degenerates for permutations with all blocks of bounded size, and our methods produce no lower bounds at all. (For the special case $b(\sigma) = n$ — no two consecutive positions carrying consecutive values — exhaustive computation for $n \leq 12$ found every such permutation to be PC-Hamiltonian, $\rho = n$; this extreme case is wide open.)

On the gadget side, the exhaustive minimum $\min_{\pi \in S_b} M(\pi) = 3$ for $4 \leq b \leq 9$ hints that transit numbers cannot be made arbitrarily small; deciding whether $M(\pi) \geq 3$ for all $|\pi| \geq 4$, or



determining the true minimum growth of M as a function of b , would calibrate how far inflation alone can go. (Heuristically it cannot beat $\sqrt{n}$ in any case: the end-visit term ties the bound to the block size, and nesting inflations doubles the visit-count bound per level.)

Two further directions. First, the analogous question for the union of two Hamiltonian *cycles*: we expect the method to adapt — the port calculus acquires two wraparound cross edges and the visit count rises accordingly — but we have not carried out the analysis and do not claim the result. Second, the Lovász-side question (Section 1.2): for the graphs $G(n, \sigma)$ of [1] built from $\sigma_{a,b}$ or from the chains of Appendix A, is the maximum number of matching edges on a path still $\Theta(n)$? The properly-colored relaxation is now known to be lossy; quantifying the loss on these explicit instances may indicate whether the matching-edge parameter $f(n)$ is linear after all.

Acknowledgments

This paper answers the main question left open in the author’s earlier note [1]. The proofs and computations were AI-assisted by Claude, under the Periapsis research structure. Because of that, I have tried to make the paper checkable without trusting either the author or the assistant. The central case analysis is stated as a complete classification (Theorem 5.4) whose every step is locally verifiable. Every computational claim is reproducible from the ancillary artifact, which includes a minimal independent verifier written directly from the definitions. And the main theorem has a second, independent proof in Appendix A.

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A A second construction: trap chains

This appendix gives an independent proof of $\rho_{\min}(n) = O(\sqrt{n})$ by a different family, found in a parallel verification effort. It does not use Sections 3–5; conversely, the main text does not use this appendix. The mechanism is again a parity obstruction, but localized differently: here *large reversed blocks of odd size, flanked by singletons*, are traps that no properly colored path can cross.

Definition A.1. Let $C = (c_0, c_1, \dots, c_r)$ be a composition of n with $c_t \geq 1$ and *no two consecutive entries equal to 1*. Put $P_t = c_0 + \dots + c_{t-1}$, and let block t occupy positions and values $[P_t, P_t + c_t - 1]$, reversed:

$$\sigma_C(P_t + i) = P_t + c_t - 1 - i, \quad 0 \leq i < c_t.$$

Write $L_t = P_t$ and $R_t = P_t + c_t - 1$ (the *ports* of block t).

Lemma A.2. H_{σ_C} consists exactly of:

- (a) doubled pairs $\{p, p + 1\}$ for all consecutive positions inside each block;
- (b) for each $0 \leq t < r$, one P-only cross edge $\{R_t, L_{t+1}\}$;
- (c) for each $0 \leq t < r$, one V-only cross edge $\{L_t, R_{t+1}\}$;
- (d) nothing else.

In particular, the cross edges incident to block t attach as follows (port–color crossover): at L_t , a P-edge toward block $t - 1$ and a V-edge toward block $t + 1$; at R_t , a V-edge toward block $t - 1$ and a P-edge toward block $t + 1$ (omitting nonexistent neighbors). Non-port vertices carry no cross edges.

Proof. (a) Inside a block, consecutive positions hold consecutive (descending) values: both a P-edge and a V-edge. (b) The positions R_t, L_{t+1} are consecutive; their values are P_t (block minimum) and $P_{t+1} + c_{t+1} - 1$ (block maximum), which differ by $c_t + c_{t+1} - 1 \geq 2$ since not both c_t, c_{t+1} equal 1: P-only. (c) The values $P_t + c_t - 1$ (maximum of block t , held at position L_t because the block descends) and P_{t+1} (minimum of block $t + 1$, held at R_{t+1}) are consecutive; the positions L_t, R_{t+1} differ by $c_t + c_{t+1} - 1 \geq 2$: V-only. (d) Positions in non-consecutive blocks are never adjacent; values in non-consecutive blocks never differ by 1 (value ranges are contiguous in block order); and within consecutive blocks, (b)/(c) list the only position-adjacent (resp. value-adjacent) pairs not internal to a block. The crossover statement reads off (b) and (c) at blocks $t - 1, t, t + 1$. $\square$

Lemma A.3. Let Q be a PC path in H_{σ_C} and let t be a block containing no endpoint of Q . Then every maximal run of consecutive Q -vertices inside block t is either (i) a single port vertex, or (ii) a full traversal of all c_t vertices from one port to the other. In case (ii), if the entering cross edge has color χ , the exiting cross edge has color $\neq \chi$ if and only if c_t is odd.

Proof. A run with no endpoint of Q has an entering and an exiting cross edge, which by Lemma A.2 attach only at ports; the run’s internal edges lie in the block, whose internal graph is a path (with doubled pairs). If the run enters and leaves at the same port it is that single vertex (re-reaching the entry port would repeat it); the two cross edges at one port have different



colors (Lemma A.2), as the PC condition requires. Otherwise it goes from one port to the other through the entire internal path: a full traversal, with $c_t - 1$ internal edges. The PC condition forces the internal colors to alternate starting with $\bar{\chi}$, so the last internal edge has color χ iff $c_t - 1$ is even; the exit edge differs from the last internal edge, so it has color $\bar{\chi}$ iff c_t is odd. $\square$

Lemma A.4 (Trap Lemma). *Let block t of σ_C have odd size $c_t \geq 3$ and suppose its neighbors are singleton blocks: $c_{t-1} = c_{t+1} = 1$ (with $1 \leq t \leq r - 1$). Let Q be a PC path with no endpoint in block t . Then Q uses at most 2 vertices of block t (at most one visit per port, and no full traversal).*

Proof. Write A for block t and let s^-, s^+ be the (single) vertices of blocks $t - 1, t + 1$. By Lemma A.2 the four cross edges at A are

$$\text{at } L_A : \{s^-, L_A\} (P), \quad \{L_A, s^+\} (V); \quad \text{at } R_A : \{s^-, R_A\} (V), \quad \{R_A, s^+\} (P).$$

Suppose Q makes a full traversal of A , entering by a cross edge of color χ ; since c_t is odd, it exits by a cross edge of color $\bar{\chi}$ (Lemma A.3). Four cases:

1. enter at L_A by $\{s^-, L_A\}$ ($\chi = P$): the exit is the V-colored cross edge at R_A , namely $\{s^-, R_A\}$ — but s^- was the vertex visited immediately before entering, a repetition;
2. enter at L_A by $\{L_A, s^+\}$ ($\chi = V$): the exit is the P-edge $\{R_A, s^+\}$ — repeats s^+ ;
3. enter at R_A by $\{s^-, R_A\}$ ($\chi = V$): the exit is the P-edge $\{s^-, L_A\}$ — repeats s^- ;
4. enter at R_A by $\{R_A, s^+\}$ ($\chi = P$): the exit is the V-edge $\{L_A, s^+\}$ — repeats s^+ .

Each case contradicts simplicity of Q . So every visit is a single port vertex; there are two ports. $\square$

Theorem A.5. *Let $L \geq 3$ be odd, $k \geq 1$, $0 \leq p \leq L + 1$, and $C = (2, (L, 1)^k, 3 + p, 2)$, a composition of $n = (L + 1)k + p + 7$ with no two consecutive 1's. Then*

$$\rho(\sigma_C) \leq 3k + 3L + p + 1.$$

Consequently $\rho_{\min}(n) \leq 7\sqrt{n} + O(1)$ for all sufficiently large n , by choosing L the largest odd integer $\leq \sqrt{n}$ and $k = \lfloor (n - 7)/(L + 1) \rfloor$, $p = n - 7 - (L + 1)k$.

Proof. The blocks, in order, are: a 2-block; $A_1, s_1, A_2, s_2, \dots, A_k, s_k$ where each A_t has size L and each s_t is a singleton; a $(3 + p)$ -block; a 2-block. For $2 \leq t \leq k$ the block A_t is flanked by the singletons s_{t-1} and s_t , so Lemma A.4 applies to it; A_1 is flanked on the left by the initial 2-block and is not trapped.

Fix a PC path Q and charge each block with the number of Q -vertices it contains; $\rho(Q)$ is the sum of the charges. We bound the charges by category:

- each singleton s_t : charge ≤ 1 (its size); total $\leq k$;
- each trapped block A_t ($2 \leq t \leq k$) containing no endpoint of Q : charge ≤ 2 (Lemma A.4); there are at least $(k - 1) - 2$ such blocks, and at most 2 trapped blocks may contain an endpoint of Q , each with charge $\leq L$ (its size). Hence the trapped blocks contribute at most $2(k - 1) + 2(L - 2)$ in total (each endpoint block exceeds the trap bound 2 by at most $L - 2$);



- A_1 : charge $\leq L$ (its size);
- the three remaining blocks: charges $\leq 2, \leq 3 + p, \leq 2$ (their sizes); total $\leq 7 + p$.

Summing: $\rho(Q) \leq k + 2(k - 1) + 2(L - 2) + L + (7 + p) = 3k + 3L + p + 1$. (If an endpoint of Q lies in a non-trapped block, the corresponding excess term is simply not needed; the bound only improves.)

For the corollary: with $L \leq \sqrt{n}$ odd and maximal, $k \leq n/(L + 1) \leq n/L$ and $p \leq L + 1$, so $\rho \leq 3n/L + 4L + 4 = 7\sqrt{n} + O(1)$ using $L = \sqrt{n} - O(1)$. $\square$

Remark A.6. The parity hypothesis is again essential: for even L , Lemma A.3 makes a full traversal *preserve* the boundary color, the four cases of Lemma A.4 become consistent, and computation shows the analogous even- L chains are fully traversable (ρ grows with slope 1). Exact computation (artifact `log check_trap.log`, plus the parallel C++ solver) gives the exact values $\rho(\sigma_C) = 2k + 2(L - 1)$ for $C = (2, (L, 1)^k, 3, 2)$, odd $L \in \{5, 7, 9, 13, 15, 21\}$ and all $k \geq 2$ within reach ($n \leq 57$): the same per-period slope 2 as the family of Section 6. We do not know whether the two families are related beyond sharing this slope.

